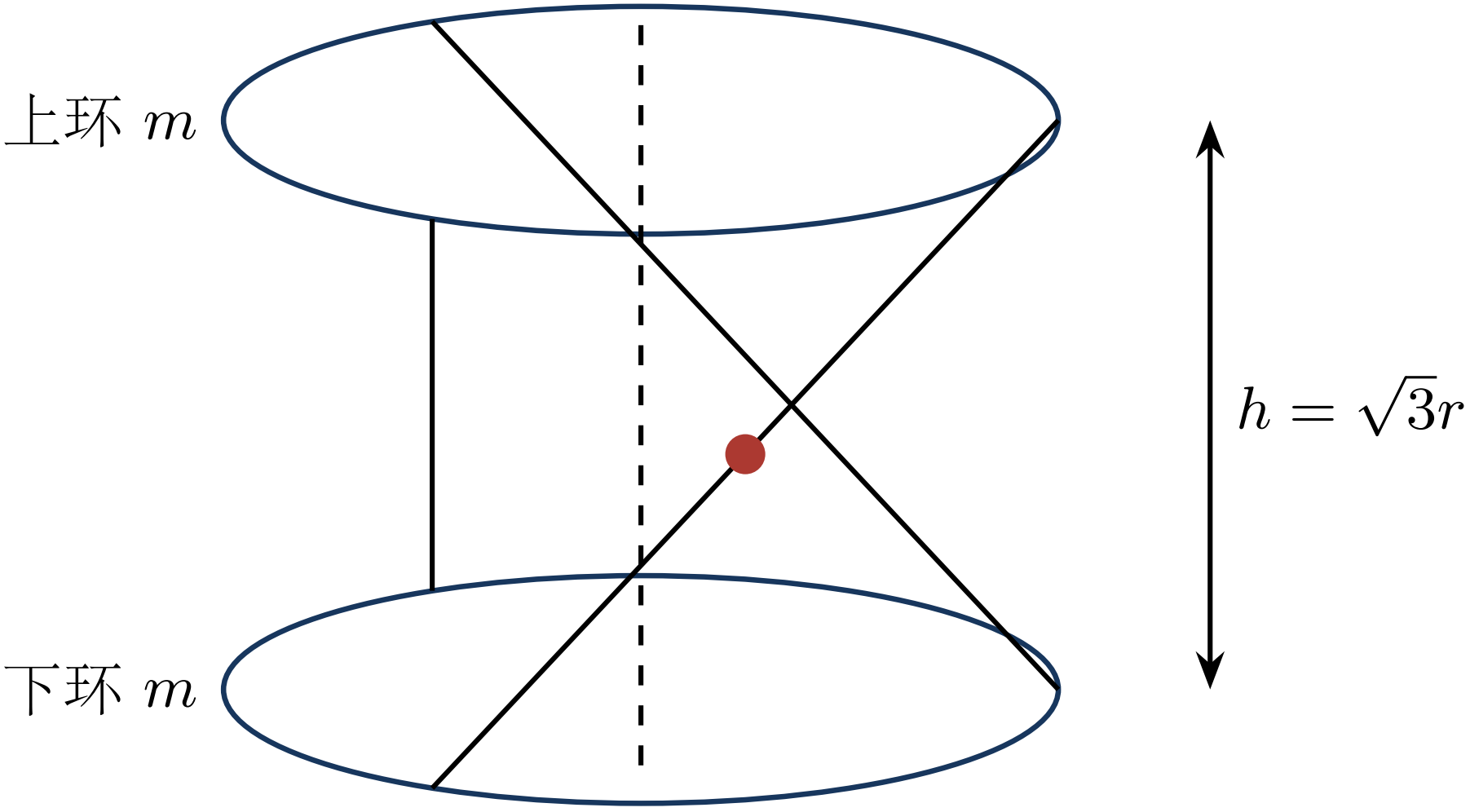
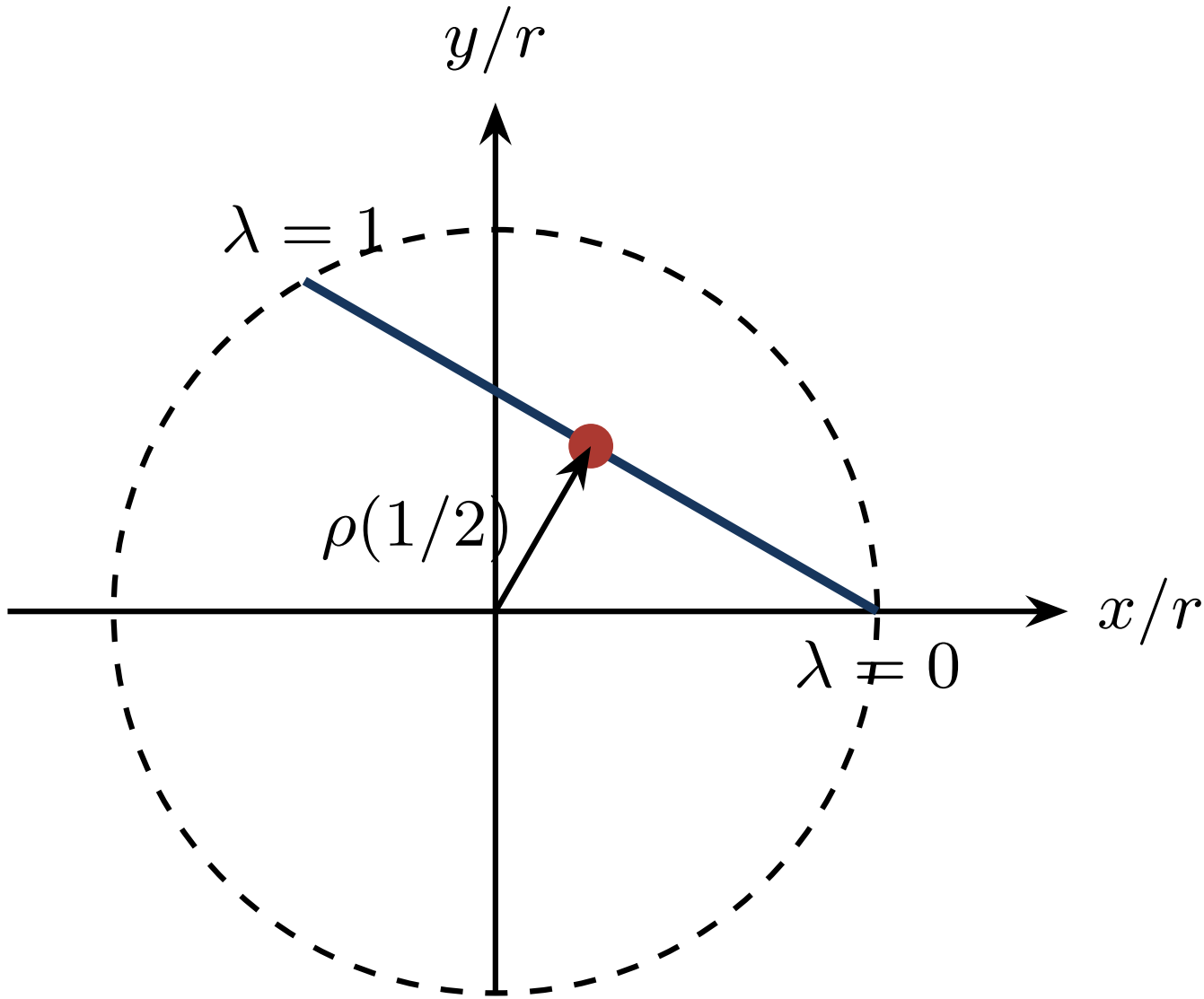


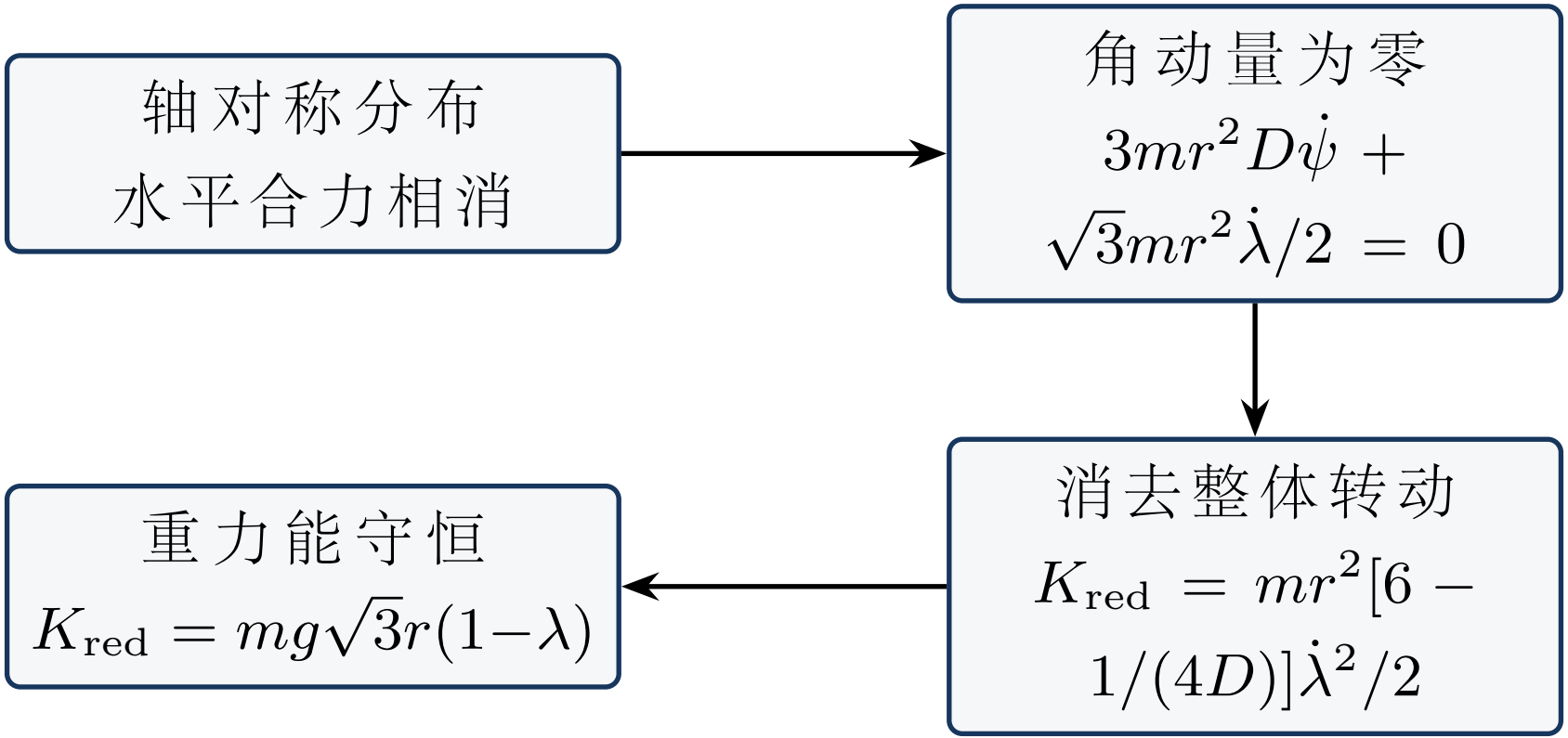
投影相差 $120°$；棒长 $\sqrt{6}r$

所有珠粒总质量 m

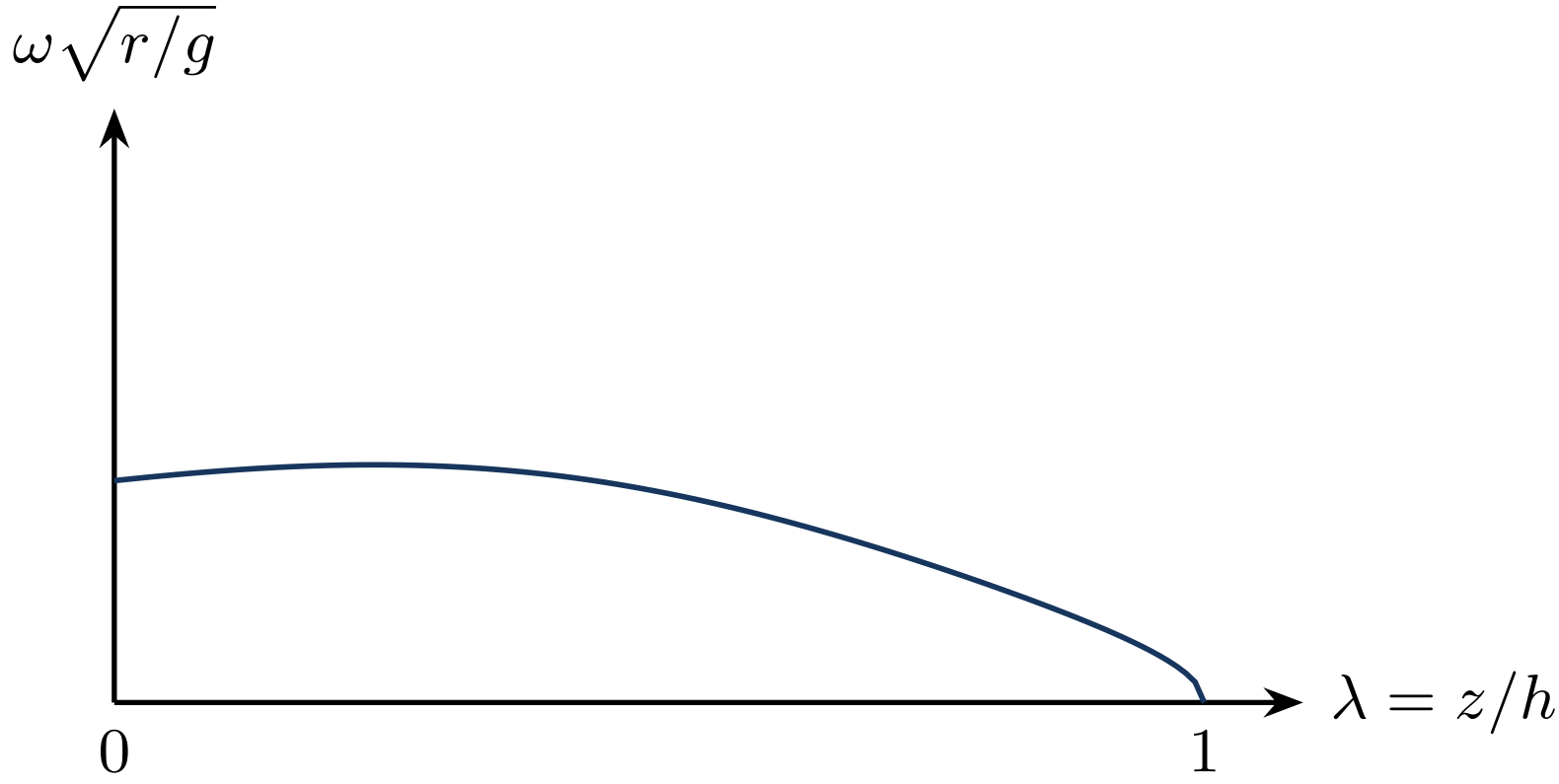


弦上位置：$\rho = r[(1-\lambda) + \lambda e^{i2\pi/3}]$

最小投影半径 $r/2$，并非所有珠粒始终在环上

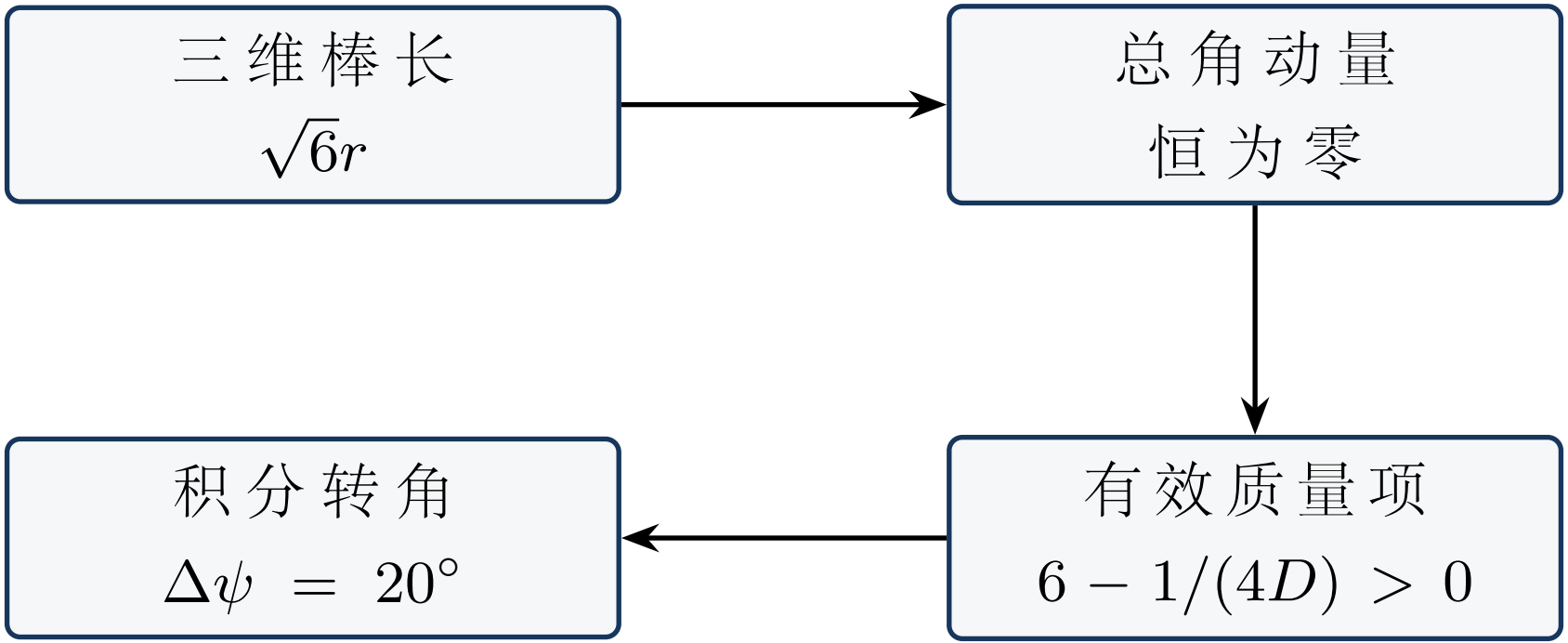


$D = \lambda^2 - \lambda + 1$；珠粒竖直速度也计入动能



从右向左下滑：上端角速度为零

中点 $\omega^2 = 4\sqrt{3}g/(153r)$，总转角 $\pi/9$



镜像棒的扭转方向，角速度与转角同时反号